APPENDIX 1

Documentation Assessment Rubrics (40 marks)

Criteria	Missing or Unacceptable (0-2)	Poor (3-4)	Accomplished (5-7)	Good (8-10)
Introduction	No or very little discussion on existing problem and the project The proposed project already exists, or with very minor change.	Little discussion on existing problem and introduction of proposed project. Minor ideas are modified from existing system(s).	Good discussion and evaluation of existing problem and the proposed project. Ideas modified from existing system, with some creative ideas are added.	A very good discussion and evaluation of existing problem and the proposed project. Majority of the ideas are creative.
Research Background (20)	Background study are retrieved directly from the literature without any paraphrasing. No discussion or very little of introduction given to the related system or technology. 0-4	Background study is lengthy, contents are retrieved directly from the literature without any critical evaluation. Introduction to the related system is given, but no evaluation provided. 5-9	Background study is concise and clear, which integrates critical and logical details from the peer-reviewed theoretical and research literature. Brief discussion and evaluation of the related system. 10-15	Background study is concise and clear, which integrates critical and logical details from the peer-reviewed theoretical and research literature. A very good discussion and evaluation of the related system. 16-20
Methodology (20)	The description does not relate the case study. Brief design of proposed methodis provided but lack of explanation or irrelevant. 0-4	Brief description of system design, with some explanations. Introduction to the relatedapplication of the methods is given but lack of examples, understanding or explanation. 5-9	System design is well-illustrated, and with clear explanation. Good discussion and evaluation of the methods applied. 10-15	System design is well-illustrated, with good explanation. Good discussion and evaluation of the relevant and practical methods appliedto the project 16-20
Results (20)	Testing methods were missing or inappropriately aligned with data and research design. Results were confusing. 0-4	Testing methods were identified but the results were confusing, incomplete or lacked relevance to the research questions, data, or research design. 5-9	The testingmethods were identified. Results were presented. All were related to the research question and design. Sufficient metric or measurement is applied. 10-15	Testing methods and results presentation were sufficient, specific, clear, structured and appropriate based on the research questions and research design. Extra metric or measurement is applied. 16-20
Discussion and Conclusion	Discussions or answers to the research objectives and results were omitted or confusing. No or very little discussion on limitation and future improvement.	Little discussions were presented. Answers to the research question andresults were unclear or confusing. Only little discussion on limitation and future improvement.	Discussions of the results were presented. The research question and system performance were answered and identified. Some discussion on limitation and future improvement were given.	The significance of the results and achievements of objectives were answered and evaluated Limitations and future improvements of the studies were identified.
Spelling, Grammar and Writing Mechanics	There were so many errors that meaning was obscured, make the content became difficult to understand Possibly copied from the source without paraphrasing.	Some grammar or spelling errors were spotted. Somesentences were awkwardly constructedand hard to understand.	There were occasional spelling or grammatical errors, but they did not represent a major distraction or obscure meaning.	Sentences were well-phrased. The writing was free or almost free ofspelling and grammatical errors.
References	No proper referencing/citation is done. Only rely on website content, but no research papers.	Reference list is provided with some mixture of reference sources including journal. Proper citation/referencing is missing.	Referencing/citation is done properly but with only little sources/reference	Rich mixture of reference sources especially good quality of research papers. Proper citations are done whenever necessary.

APPENDIX 2

Prototype Assessment Rubrics (60 marks)

No	Item	Criteria		
		Poor	Accomplished	Good
1	User interface / output (20)	Poor or confusing design of UI or output, which provides inadequate information/outputs Most of the information/outputs generated are less accurate. Layout of information is not organized. 0-8	Adequate information/outputs needed are generated The information/output generated are accurate but some with errors. Layout of information is organized. 9-15	All the necessary information/outputs are generated All or most of the information/outputs generated are accurate. Minor errors can be ignored. Layout of information is well-organized. 16-20
2	Programming (30)	The end product fails with many logic errors, many actions lacked exception handling. Solutions are over-simplified. Programming skill needs improvement. Minimal validations are provided. Business rules are not validated 0-10	Major parts are logical, but some steps to complete a specific job may be tedious or unnecessarily complicated. Program algorithm demonstrates acceptable level of complexity. The student is qualified to be a programmer. Important and necessary validations are provided 11-19	Correct and logical flow, exceptions are handled well. Demonstrates appropriate or high level of complex algorithms and programming skills. Thorough and thoughtful validations are provided. All important business rules are validated 20-30
3	Degree of completion (20)	Too much still remain to be done. Basic requirements are not fulfilled. The end product produces enormous errors, faults or incorrect results 0-8	All required features present in the interface within the required scope, but some are simplified. Or one or two features are missing. The system is able to run with minor errors 9-15	All required features present in the interface within or beyond the required scope no bugs during demonstration 16-20
4	System implementation (20)	The end product is produced with different system design or approach, which is not related to the initial proposal 0-8	The end product conforms to most of the system design, but some are different from the specification 9-15	The end product fully conforms to the proposed system design 16-20
5	On-the-spot coding (10)	The student is unclear about the work produced, sometimes not even knowing where to find the source code. 0-4	The student knows the code whereabouts, but sometimes may not be clear why the work was done in such a way. 5-7	The student is clear about every piece of the work done. 8-10